Rational Pairs Suffice for Illumination in Orthogonal Lattice Rooms

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Abstract

Let P be a strictly simple orthogonal polygon with integer vertices. We prove that if every pair of distinct rational interior points illuminates, then every pair of distinct real interior points illuminates. More precisely, any dark pair with an irrational coordinate belongs locally to a one-parameter family of dark pairs containing rational pairs arbitrarily nearby. The proof uses the four-copy square-tiled unfolding, horizontal and vertical multi-twists, and elementary density arguments on circles and tori. The reduction gives no bound on rational denominators and requires no limiting deformation of the room.

1 Statement and conventions

A room is the closed region bounded by a Jordan polygonal curve. Throughout, P is orthogonal, has integer vertices, and has no holes, self-contacts, or interior mirrors. A legal billiard path has finitely many straight legs, reflects specularly at smooth boundary points, and never meets a polygon vertex. Distinct interior points *illuminate* one another if a legal path connects them; otherwise the pair is *dark*. A point is rational when both coordinates are rational.

Theorem 1 (Rational-pair reduction)**.** *If every pair of distinct rational points in* $\mathrm{int}(P)$ *illuminates, then every pair of distinct points in* $\mathrm{int}(P)$ *illuminates.*

We prove the following stronger local statement.

Theorem 2 (A dark irrational pair slides)**.** *Suppose* (S, T) *is a dark pair of distinct interior points and* S *is not rational. After exchanging the coordinate axes if necessary, there are* $x \in \mathbb{R}\setminus\mathbb{Q}$, $y, v, c \in \mathbb{Q}$, *and* $\rho \in \mathbb{Q} \setminus \{0\}$ *such that*

$$S = (x, y), \qquad T = (\rho x + c, v).$$

For all sufficiently small real δ*, the distinct interior pair*

$$S_\delta = (x + \delta, y), \qquad T_\delta = (\rho x + c + \rho\delta, v)$$

is dark. In particular, there are rational dark pairs arbitrarily close to (S, T)*.*

The result concerns a fixed room. It does not say that dark pairs are rational, that their number is finite, or that a bounded-denominator grid can certify illumination. No wall-count bound is imposed.

2 The marked four-copy unfolding

Let $K = \{1, -1\}^2$. Form M from four copies of P labelled by $e = (e_1, e_2) \in K$, with developing charts $(e_1 x, e_2 y)$. Glue vertical boundary edges to the sheet with e_1 reversed, and horizontal boundary edges to the sheet with e_2 reversed. The resulting connected translation surface is tiled by unit squares. The map

$$\pi : M \longrightarrow \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2, \qquad \pi(S, e) = (e_1 x, e_2 y) \pmod{\mathbb{Z}^2}$$

is a finite covering away from 0. Put $L = \pi^{-1}(0)$ and let $\Sigma \subset L$ consist of all lifts of polygon vertices. The lifts of convex vertices are regular metric points, but remain *marked forbidden points*. Smooth boundary lattice points and interior lattice points are not forbidden.

Call a pair of distinct points in $M \setminus \Sigma$ *surface-lit* if a straight geodesic segment joins them while avoiding Σ. For distinct room points S, T, fix $s = (S, (1, 1))$ and write $t_e = (T, e)$. Then

$$(S, T) \text{ illuminates} \quad \Longleftrightarrow \quad (s, t_e) \text{ is surface-lit for some } e \in K. \tag{1}$$

Indeed, reflect each successive billiard leg into the adjacent sheet; conversely project a geodesic through the sheet identifications. Marking all vertex lifts makes corner avoidance identical in both directions. A compact geodesic avoiding the finite marked set has only finitely many crossings of the finite polygon edges. For interior endpoints, a segment cannot run along a boundary edge. This is the usual unfolding correspondence; see also [1, Lemma 2.1 and Remark 2.2].

Lemma 1 (Openness)**.** *The surface-lit pairs form an open subset of* $(M \setminus \Sigma)^2$ *minus the diagonal.*

Proof. A legal segment is compact and has positive distance from Σ. Continue it slightly at both ends and develop a sufficiently thin strip around it by parallel transport. The strip maps locally isometrically to $M \setminus \Sigma$; global injectivity is unnecessary. Small changes of the two endpoints in this strip are joined by a straight segment remaining inside the strip. Its image is a legal geodesic. The endpoint charts are local homeomorphisms, proving openness. □

2.1 Two multi-twists

Cut M along all horizontal grid edges. Each component is a horizontal row cylinder of height 1 and integer circumference. Choose a positive integer A divisible by every such circumference. On a cylinder with coordinates $(u, \eta) \in (\mathbb{R}/\ell\mathbb{Z}) \times [0, 1]$, set

$$h(u, \eta) = (u + A\eta, \eta).$$

The boundary values are the identity, so the maps glue to an affine homeomorphism of M fixing every horizontal grid edge pointwise. Similarly there is a vertical multi-twist v, with a positive integer parameter B, fixing every vertical grid edge pointwise. Their derivatives and torus actions are

$$Dh = \begin{pmatrix} 1 & A \\ 0 & 1 \end{pmatrix} = H, \qquad Dv = \begin{pmatrix} 1 & 0 \\ B & 1 \end{pmatrix} = V, \qquad \pi h = H\pi, \quad \pi v = V\pi.$$

Both fix L, and hence Σ, pointwise. Every element of $G = \langle h, v \rangle$ preserves surface illumination: it maps straight segments to straight segments and preserves the forbidden set. The diagonal action of G on $M \times M$ will be used below. We do not assume that this action descends to pairs of room points.

3 Density under horizontal and vertical twists

We use the elementary torus-rotation fact that multiples of a vector $(\alpha_1, \ldots, \alpha_d)$ are dense modulo a period lattice when $1, \alpha_1, \ldots, \alpha_d$ are rationally independent after normalizing the periods. For a circle this is the irrational-rotation theorem; for a product of circles it is Kronecker's theorem.

Lemma 2 (One-point density)**.** *Let X be a compact connected square-tiled translation surface, with tile side $1/N$ and tiling map to $\mathbb{T}^2$. Suppose affine maps h_0, v_0 fix all tile vertices and have derivatives $\left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ b & 1 \end{smallmatrix}\right)$, where a, b are nonzero integers. Every point with an irrational torus coordinate has dense $\langle h_0, v_0 \rangle$-orbit.*

Proof. First pass to positive powers that fix every horizontal, respectively vertical, tile edge pointwise. Such powers exist: there are finitely many edges, each is mapped to an edge of the same direction and length, and an edge preserved with its orientation is fixed pointwise. Fixing vertices alone would not suffice, since different prongs can be permuted. The derivatives of these powers still have nonzero integral shear parameters. It suffices to prove density for the resulting subgroup.

Suppose the second torus coordinate is irrational. The point lies in an open row cylinder at an irrational transverse height. The horizontal twist acts on its horizontal leaf by an irrational rotation, because the circumference and tile offsets are rational. Thus the orbit closure Y contains that entire leaf. It crosses each tile it meets horizontally. In any column cylinder met by the leaf, points on the leaf with irrational first coordinate have dense vertical-twist orbits on their vertical leaves. These leaves are dense in that column cylinder, so its closure lies in Y.

Call a row or column cylinder full if its closure lies in Y. If a full cylinder and one of the other direction share a tile, the second is full: use the irrational transverse coordinates in their common tile and the corresponding twist. Every pair of edge-adjacent tiles belongs to a common row or column cylinder. Connectedness propagates fullness through every tile. If only the first coordinate is irrational, interchange the axes. □

Lemma 3 (Mixed rationality)**.** *If $\pi(s)$ is rational and $\pi(t)$ is not, then (s, t) is surface-lit.*

Proof. The G-orbit of s is finite: its torus coordinates have a fixed bounded denominator, and π has finite fibres. Choose $n > 0$ such that both h^n and v^n fix s. Lemma 2 makes the orbit of t under $\langle h^n, v^n \rangle$ dense. Move it into a sufficiently small flat disc around the regular point s, disjoint from Σ. The distinct endpoints then join by a segment. Apply the inverse affine map. □

A pair $(z, w) \in \mathbb{T}^2 \times \mathbb{T}^2$ is *generic* here if no nonzero $(a, b) \in \mathbb{Z}^2$ satisfies $az + bw \in \mathbb{Q}^2/\mathbb{Z}^2$.

Lemma 4 (Generic-pair density)**.** *If $(\pi(s), \pi(t))$ is generic, the diagonal G-orbit of (s, t) is dense in $M \times M$.*

Proof. Write $\pi(s) = (x, y)$ and $\pi(t) = (u, w)$, and let Y be the orbit closure. Let r_x be the rank of x, u in the vector space $\mathbb{R}/\mathbb{Q}$ over $\mathbb{Q}$, and r_y the rank of y, w.

If $r_y = 2$, then $1, y, w$ are rationally independent. Horizontal twisting fills the product of the horizontal leaves through s, t: their circumferences are integers, so period normalization preserves the independence. On this product, points with $1, x', u'$ rationally independent are dense. Vertical twisting fills their products of vertical leaves. Hence Y contains the product of the closures of every pair of column cylinders met by those horizontal leaves.

A pair of cylinders of the same direction is full if the product of its closures lies in Y. If (C_1, C_2) is full and each C_i' intersects C_i in a tile, with the opposite direction, then (C_1', C_2') is full: apply the same

independent-rotation argument in the product of the two intersection tiles. This propagates to every tile pair. To move one tile factor while holding the other, alternate horizontal and vertical cylinder steps; in the stationary factor use its row and column through its fixed tile. A step can be padded by staying in that tile, so both factors can follow the same alternating schedule. Connectedness of the tile adjacency graph therefore gives $Y = M \times M$. The case $r_x = 2$ follows by exchanging axes.

The only remaining generic case is $r_x = r_y = 1$. Write

$$(x, u) = q' + \xi\beta, \qquad (y, w) = q + \eta\alpha,$$

with $q, q', \alpha, \beta \in \mathbb{Q}^2$, $\alpha, \beta \neq 0$, and ξ, η irrational. Genericity forces $\det(\alpha, \beta) \neq 0$, since otherwise one nonzero rational row vector annihilates both irrational parts. After taking a multiple to kill the rational offsets, horizontal twisting has a circle orbit closure in direction α. Consequently Y contains pairs obtained by moving the first and second points horizontally by $r\alpha_1, r\alpha_2$, respectively, for every sufficiently small real r (and by continuation along the leaves for all r). Choose such r with $1, \xi, r$ rationally independent. A rational relation between the new first coordinates would annihilate both β and α, which is impossible. The moved pair has $r_x = 2$ and its orbit closure is $M \times M$. If one original rank is zero, genericity forces the other to be two, already treated. □

Corollary 1. *Every surface-dark pair (s, t) satisfies $a\pi(s) + b\pi(t) \in \mathbb{Q}^2/\mathbb{Z}^2$ for some nonzero integer pair (a, b). If both torus coordinates are irrational, then $ab \neq 0$.*

Proof. Surface-lit pairs form a nonempty open invariant set. A dense orbit must meet that set. If $a = 0$ or $b = 0$, the corresponding coordinate is rational. □

4 Rational relation surfaces

Put $M^\circ = M \setminus L$.

Lemma 5 (A dark relation component)**.** *Suppose $z = \pi(s)$ is irrational, a, b are relatively prime nonzero integers, and $az + b\pi(t) = c \in \mathbb{Q}^2/\mathbb{Z}^2$. Let F_0 be the connected component through (s, t) of*

$$F = \{(s', t') \in M^\circ \times M^\circ : a\pi(s') + b\pi(t') = c\}.$$

If (s, t) is surface-dark, every distinct pair in F_0 is surface-dark.

Proof. Choose rational $z_0, w_0 \in \mathbb{T}^2$ with $az_0 + bw_0 = c$. Since $\gcd(a, b) = 1$, the solution torus downstairs is parametrized bijectively by

$$z' = z_0 + b\zeta, \qquad w' = w_0 - a\zeta, \qquad \zeta \in \mathbb{T}^2. \tag{2}$$

For example, Bezout coefficients give an inverse to the map $\zeta \mapsto (b\zeta, -a\zeta)$. Remove the finite rational set E where z' or w' is zero. The pullback F is a finite covering of $\mathbb{T}^2 \setminus E$. Completing each connected component adds finitely many points and produces a compact translation surface branched only over rational points. After subdividing a rational grid, it is tiled by squares of side $1/N$. The parameter ζ of (s, t) is irrational because z is irrational.

A power of the diagonal action $h \times h$ preserves the rational relation and the component F_0: the orbit of c on its finite rational torus grid is finite, and there are finitely many components. In the chart ζ its action is affine with derivative a power of H and rational translation part. Taking another power makes this translation part zero modulo $\mathbb{Z}^2$ and fixes every vertex of a sufficiently fine rational tile grid upstairs. To justify the first step, the affine action permutes a finite rational grid containing 0;

a power fixes 0. To justify the second, its action on the finite set of preimages of the grid has finite order. The same construction applies to $v \times v$ and V. When applying Lemma 2, take further powers to fix tile edges pointwise as well. That lemma makes the orbit of (s, t) under the resulting subgroup dense in the completed component, and hence dense in F_0.

If a distinct pair in F_0 were surface-lit, its open illumination neighbourhood would meet this orbit. Affine invariance would make (s, t) surface-lit, a contradiction. □

5 Proof of the reduction

Proof of Theorem 2. Write $S = (x, y)$, $T = (u, w)$, $s = (S, (1, 1))$, and $t_e = (T, e)$. All four pairs (s, t_e) are surface-dark by (1). If T were rational, Lemma 3 would contradict darkness. Thus both S and T are irrational points.

For each e the relation corollary supplies nonzero integers a_e, b_e with

$$a_e(x, y) + b_e(e_1 u, e_2 w) \in \mathbb{Q}^2. \tag{3}$$

Divide by their greatest common divisor to make them relatively prime; this preserves rationality of the right-hand side. Suppose both x and y were irrational. From $e = (1, 1)$ we obtain $u = rx$ and $w = ry$ modulo $\mathbb{Q}$, for nonzero rational r. From $e = (-1, 1)$ we obtain $u = r'x$ and $w = -r'y$ modulo $\mathbb{Q}$. Then $(r - r')x$ and $(r + r')y$ are rational, forcing $r = r'$ and $r = -r'$, a contradiction. Exactly one coordinate of S is irrational. Exchange axes so that it is x. The relation for $e = (1, 1)$ now gives $y, w \in \mathbb{Q}$ and $u = \rho x + c$, with $\rho \in \mathbb{Q} \setminus \{0\}$ and $c \in \mathbb{Q}$. Set $v = w$ to match the statement.

For any sheet e, the first coordinate of (3) implies $a_e + e_1 b_e \rho = 0$, since x is irrational. Therefore the path

$$\delta \longmapsto (s_\delta, t_{e,\delta})$$

obtained from the stated room slides keeps its rational relation constant. For sufficiently small δ it remains in $M^\circ \times M^\circ$, with distinct interior room endpoints. It consequently remains in the same relation component. Lemma 5 makes all four lift pairs surface-dark, so (1) gives darkness in the room.

For an explicit permissible neighbourhood, take $|\delta|$ smaller than

$$\min\left\{\operatorname{dist}(x, \mathbb{Z}), \frac{\operatorname{dist}(u, \mathbb{Z})}{|\rho|}, \operatorname{dist}(S, \partial P), \frac{\operatorname{dist}(T, \partial P)}{|\rho|}, \frac{|S - T|}{1 + |\rho|}\right\}.$$

Every entry is positive. Choosing $x + \delta$ rational gives rational coordinates for both endpoints and can be done arbitrarily close to zero. □

Proof of Theorem 1. If a dark distinct interior pair exists and both endpoints are rational, it already contradicts the hypothesis. Otherwise name an irrational endpoint S and apply Theorem 2 to obtain a rational dark pair. □

Corollary 2 (Scaling formulation)**.** *Fix a class of strictly simple orthogonal lattice rooms closed under positive integer dilations. If every pair of distinct interior lattice points illuminates in every room of the class, then every distinct real interior pair illuminates in every room of the class.*

Proof. In a fixed room, any rational pair becomes a lattice pair after a common integer dilation. Dilation preserves legal billiard paths. Apply Theorem 1. □

6 Scope and verification

The argument is analytic and concerns all real coordinates. It does not sample the endpoints, enumerate rooms, or impose a reflection cutoff. The scaling corollary does not convert an area-bounded census into an all-real theorem: scaling can exceed every given area bound. Similarly the proof supplies no universal finite list of directions, reflection words, or rational denominators.

All convex vertex lifts remain forbidden throughout. Interior lattice lifts and smooth wall lattice lifts are regular, allowed points. The affine maps used in the proof fix every lattice lift, so they preserve the complete forbidden set. Their action is used on the unfolding rather than assumed to act on room pairs.

The source reduction has independent AI mathematical review. This manuscript incorporates the review's finite-power corrections and the explicit small-slide bound. The folder provenance identifies the source and review versions. No computer calculation is needed to establish the quantified theorem.

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References

[1] A. Wolecki, *Illumination in Rational Billiards*, `arXiv:1905.09358v3` (2024), Lemma 2.1 and Remark 2.2. https://arxiv.org/abs/1905.09358.